

Quiz 5 (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Determinants

$$\begin{bmatrix} 2 & 1 & 4 \\ 1 & 2 & -1 \\ 4 & 1 & 2 \end{bmatrix}$$

Determine the determinant of the above matrix using the procedure we discussed in lecture.

(1 swap) $\begin{bmatrix} 1 & 2 & -1 \\ 2^{-2} & 1^{-4} & 4^{+2} \\ 4^{-4} & 1^{-2} & 2^{+4} \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 6 \\ 0 & -7 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & -7 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & -8 \end{bmatrix}$

(scale $\frac{1}{3}$)

$$\det(A) = \frac{(-1)^5}{c} u_{11} u_{22} u_{33} = \frac{(-1)^1}{(-1/3)} (1) (1) (-8) = -24$$

Solution.

$$-24$$

2 Characteristic Equation

Determine a matrix in $\mathbb{R}^{2 \times 2}$ with nonzero entries whose characteristic polynomial is $\lambda^2 + 2\lambda + 2$.

$$\lambda^2 + 2\lambda + 2 = (\lambda^2 + 2\lambda + 1) + 1 = (\lambda + 1)^2 + 1$$

$$\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

Note: There are multiple answers

Solution.

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

3 Eigenbases

$$\begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix}$$

Determine all eigenvalues of the above matrix, along with bases for the corresponding eigenspaces.

$$\begin{aligned} \det(A - \lambda I) &= (11 - \lambda)(-10 - \lambda) + 108 \\ &= (\lambda - 11)(\lambda + 10) + 108 & \lambda = 2, -1 \\ &= \lambda^2 - \lambda - 110 + 108 \\ &= \lambda^2 - \lambda - 2 \\ &= (\lambda - 2)(\lambda + 1) \end{aligned}$$

$$A - 2I = \begin{bmatrix} 9 & -6 \\ 18 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -2/3 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 2/3 x_2 \\ x_2 &\text{ is free} \end{aligned}$$

$$v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$A + I = \begin{bmatrix} 12 & -6 \\ 18 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -1/2 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 1/2 x_2 \\ x_2 &\text{ is free} \end{aligned}$$

$$v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Solution.

$$\lambda_1 = 2 \quad \text{Nul}(A - 2I) = \text{span} \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

$$\lambda_2 = -1 \quad \text{Nul}(A + I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$